Mansoura University
Faculty of Computers and Information
Department of Information System
First Semester- 2024-2025

Agile Methods

Assoc.Prof: Samir Abdelrazek

Lecture 11
Object Modeling (O-O-A)
Class Diagram

Assoc. Prof: Samir Abd ElRazek

2024

Documentation Of Data and Process Models

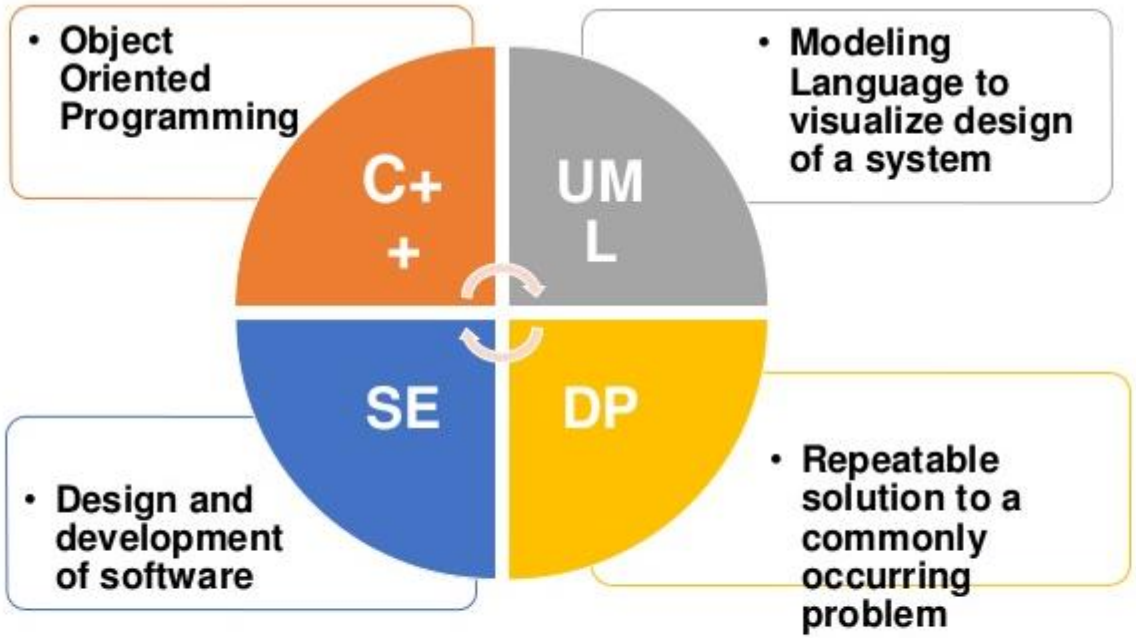
Overview of Object-Oriented Analysis

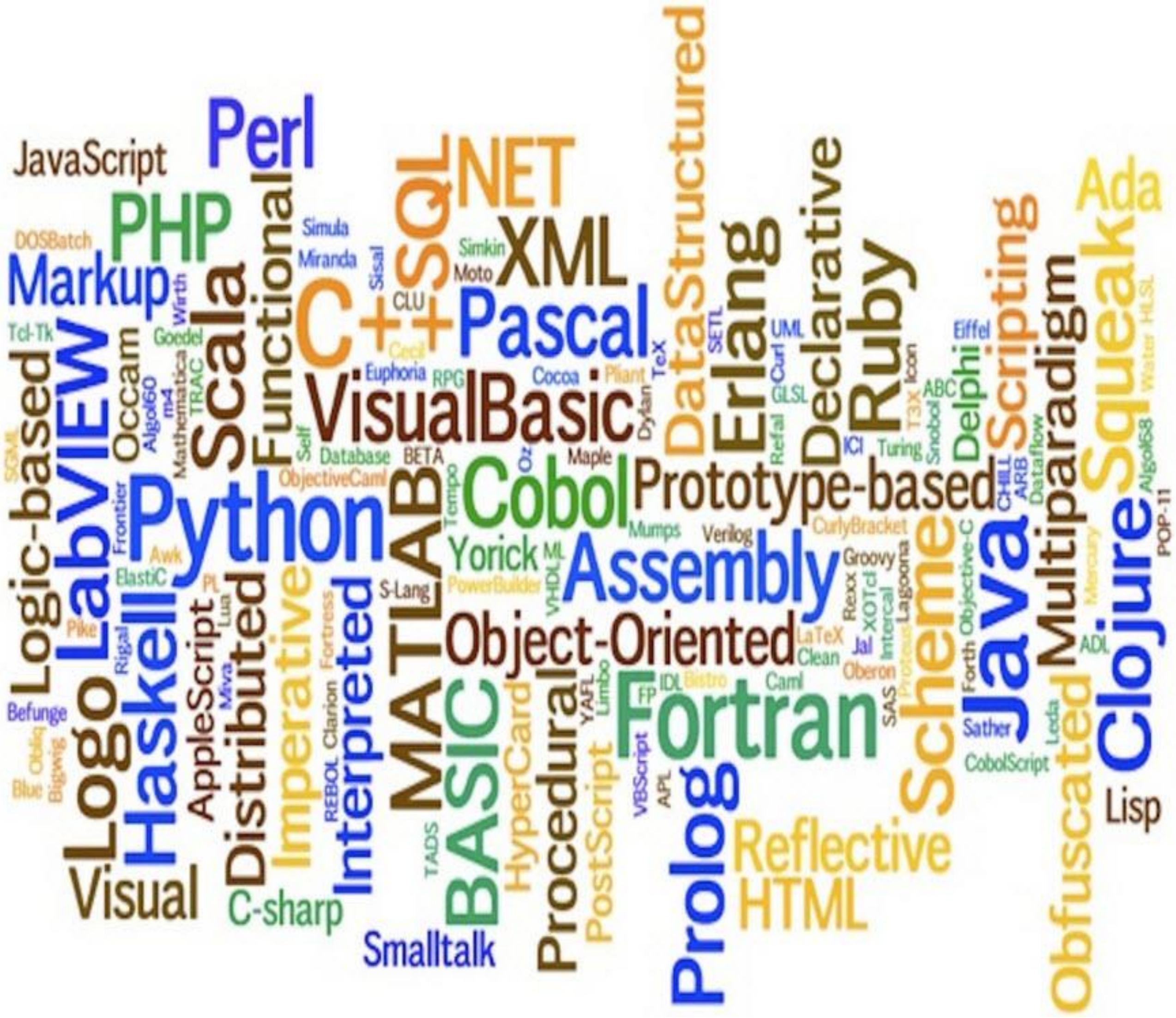
Class Diagram

- Objects
- Attributes
- Methods
- Messages
- Classes

Relationship among objects and classes

- Object Relationship diagram



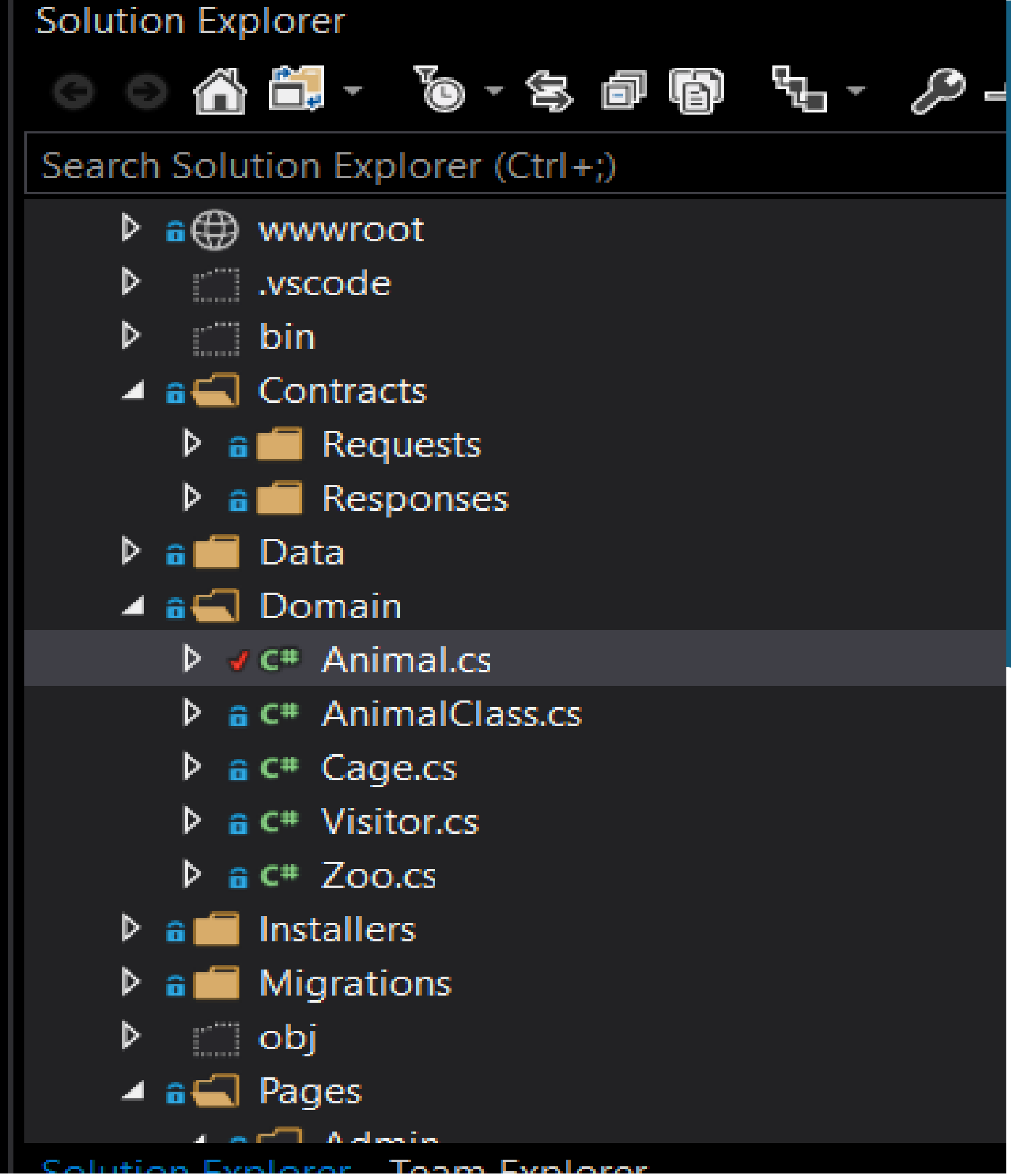


O-O Methodology

As stated in Chapter 1, the most popular systems development options are structured analysis, object-oriented (O-O) analysis, and agile methods, also called adaptive methods.

O-O methodology is popular because it integrates easily with O-O programming languages such as C# , Java , VB.Net, Python, and Dart.

Programmers also like O-O code because it is modular, reusable, and easy to maintain.

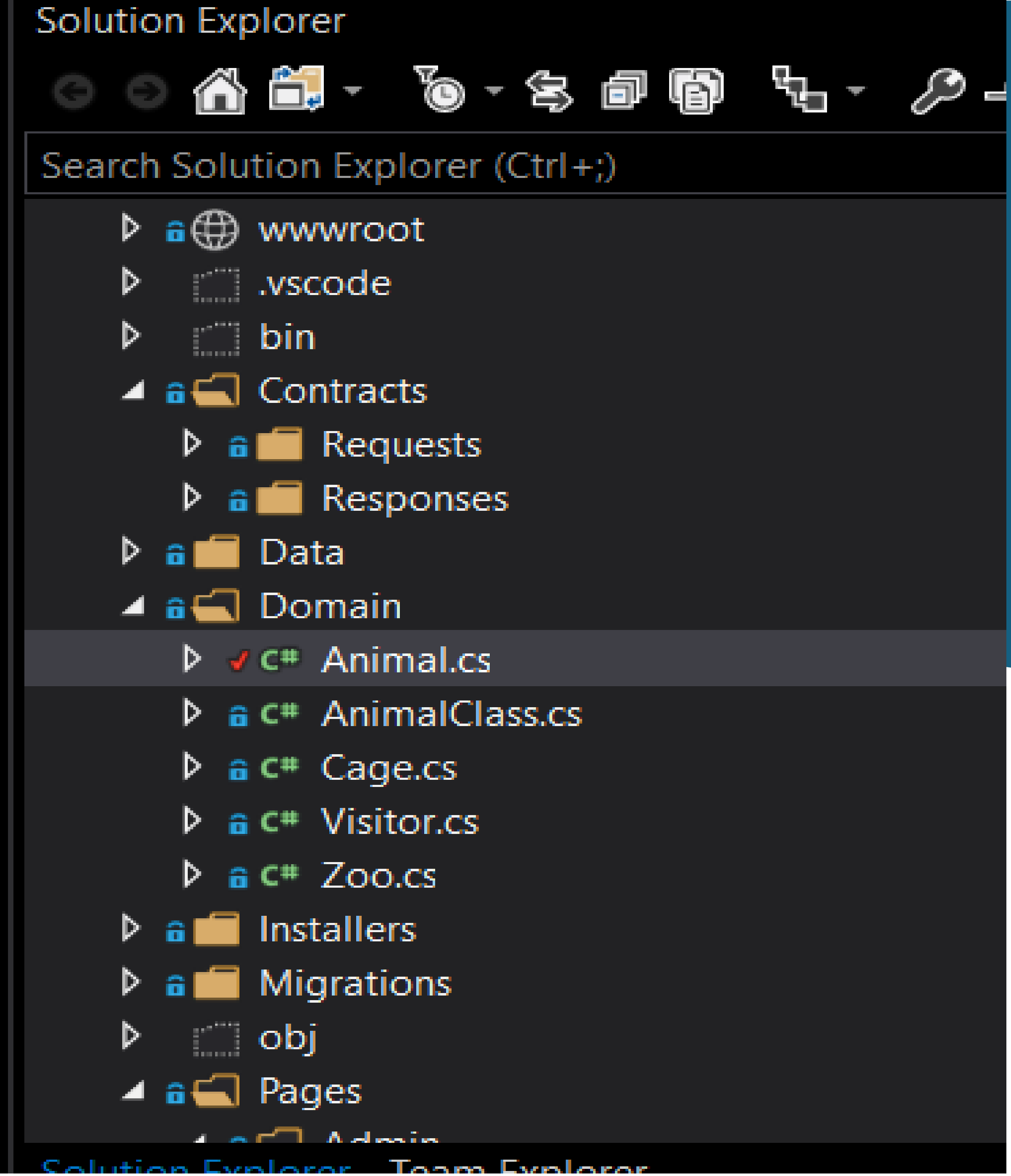


O-O Methodology

Object-oriented (O-O) analysis describes an information system by identifying things called objects.

An object represents a real person, a place, an event, or a transaction.

when a zoo have many classes of Animals and Visitors come to see the Animals, the Visitor is an object, the Animal is an object, and the Zoo itself is an object.

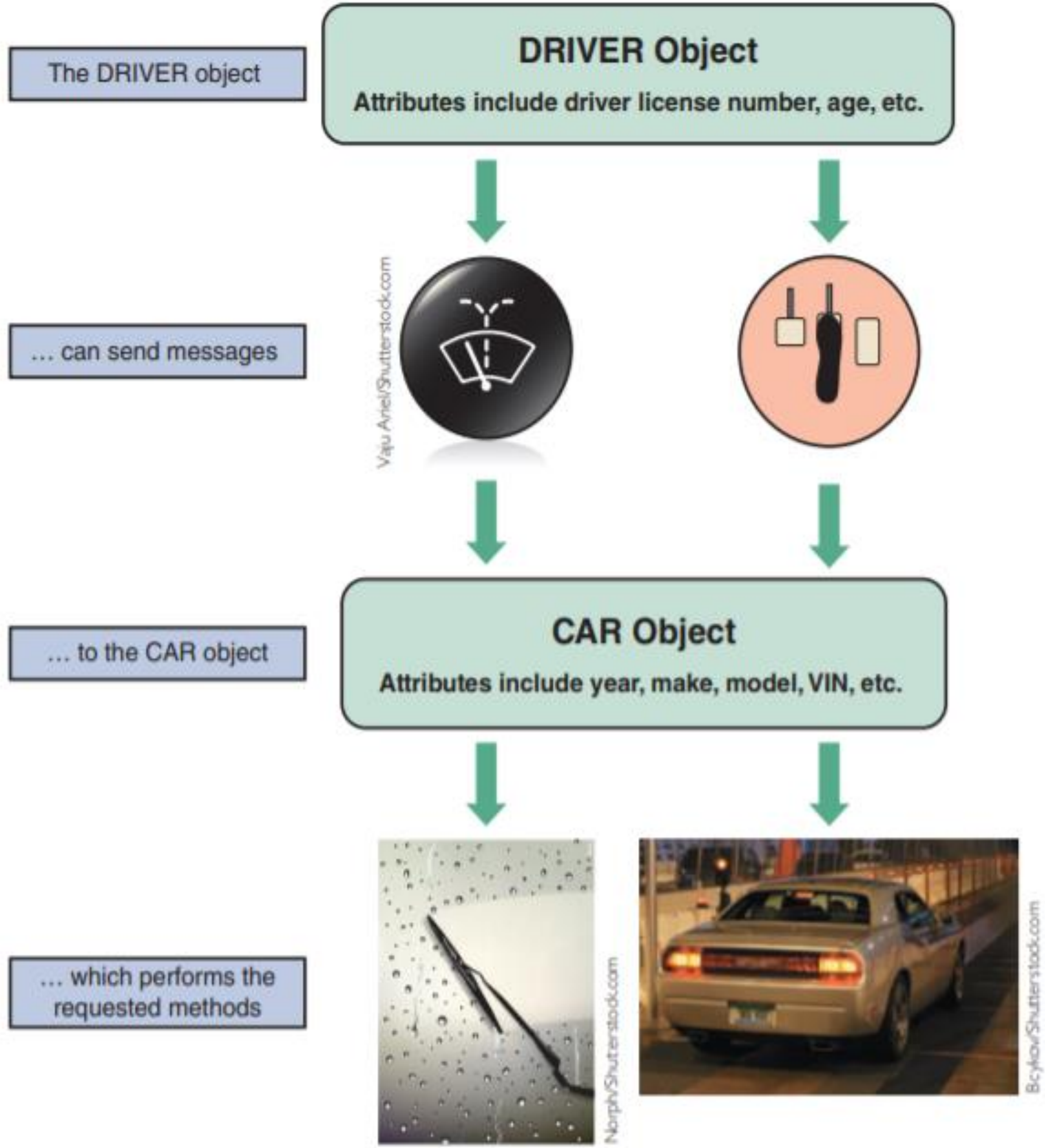


O-O Methodology

Object-oriented analysis is a popular approach that sees a system from the viewpoint of the objects themselves as they function and interact.

The end product of O-O analysis is an object model, which represents the information system in terms of objects and O-O concepts.

A collection of object Models can represent the Domain such zoo example.

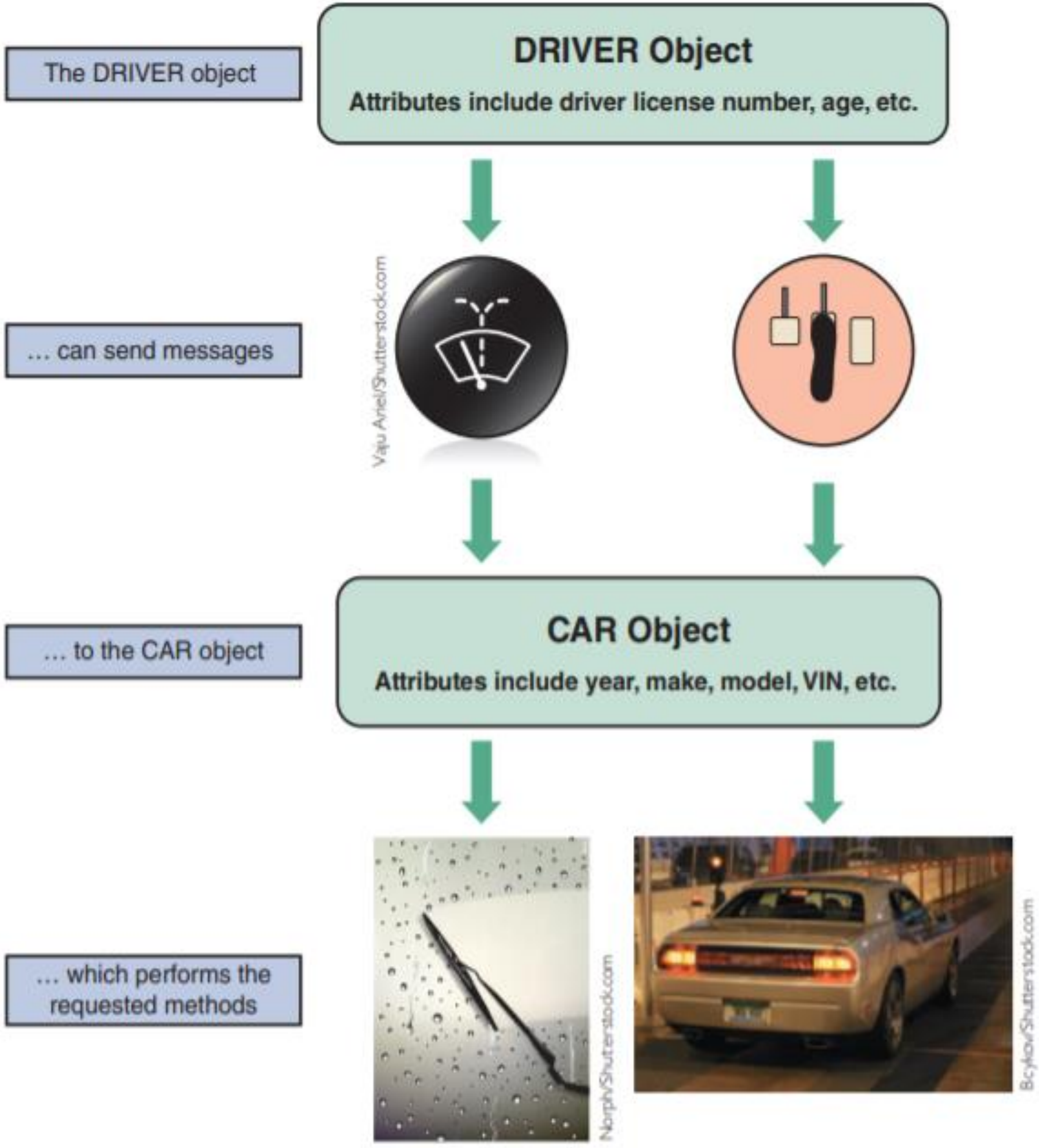


OVERVIEW OF OBJECT-ORIENTED ANALYSIS

DFDs were created that treated data and processes separately.

An object, however, includes data and the processes that affect that data.

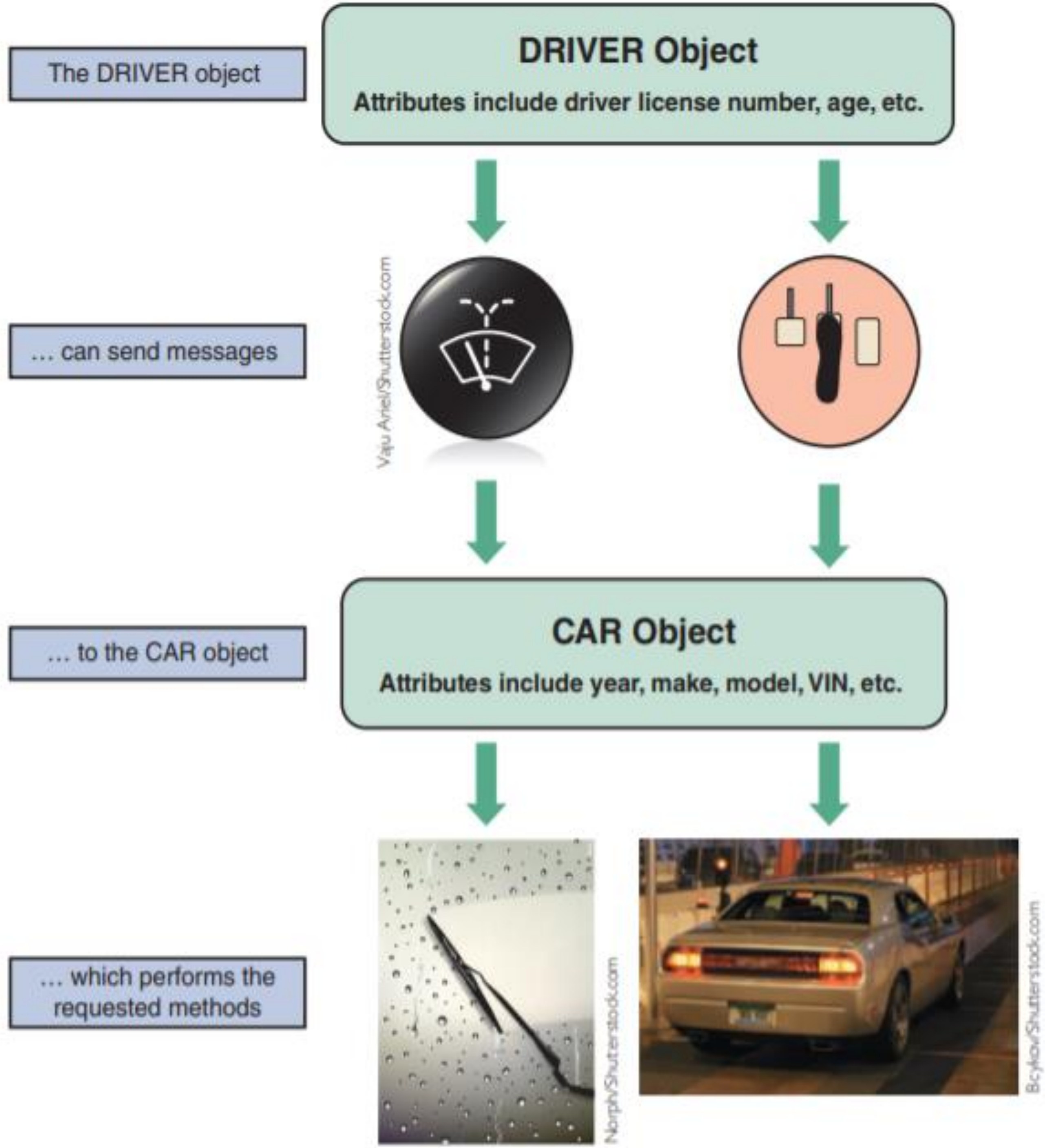
An object has certain attributes, which are characteristics that describe the object. For example, a car has attributes such as make, model, and color.



OVERVIEW OF OBJECT-ORIENTED ANALYSIS

An object also has methods, which are tasks or functions that the object performs when it receives a message, or command, to do so.

For example, a car performs a method called OPERATE WIPERS when it is sent a message with the wiper control, and it can APPLY BRAKES when it is sent a message by pressing the brake pedal.



OVERVIEW OF OBJECT-ORIENTED ANALYSIS

A class is a group of similar objects.

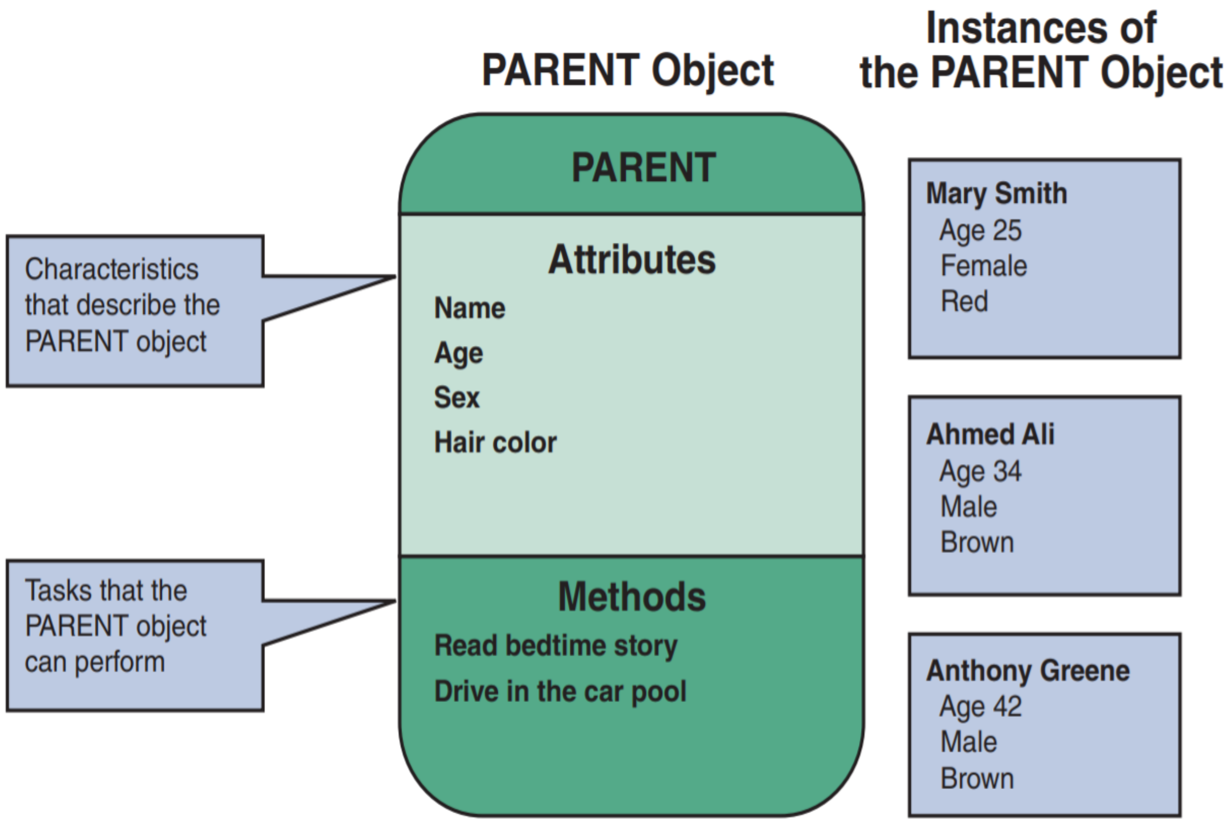
For example, Ford Fiestas belong to a class called CAR.

An instance is a specific member of a class.

A Toyota Camry, for example, is an instance of the CAR class.

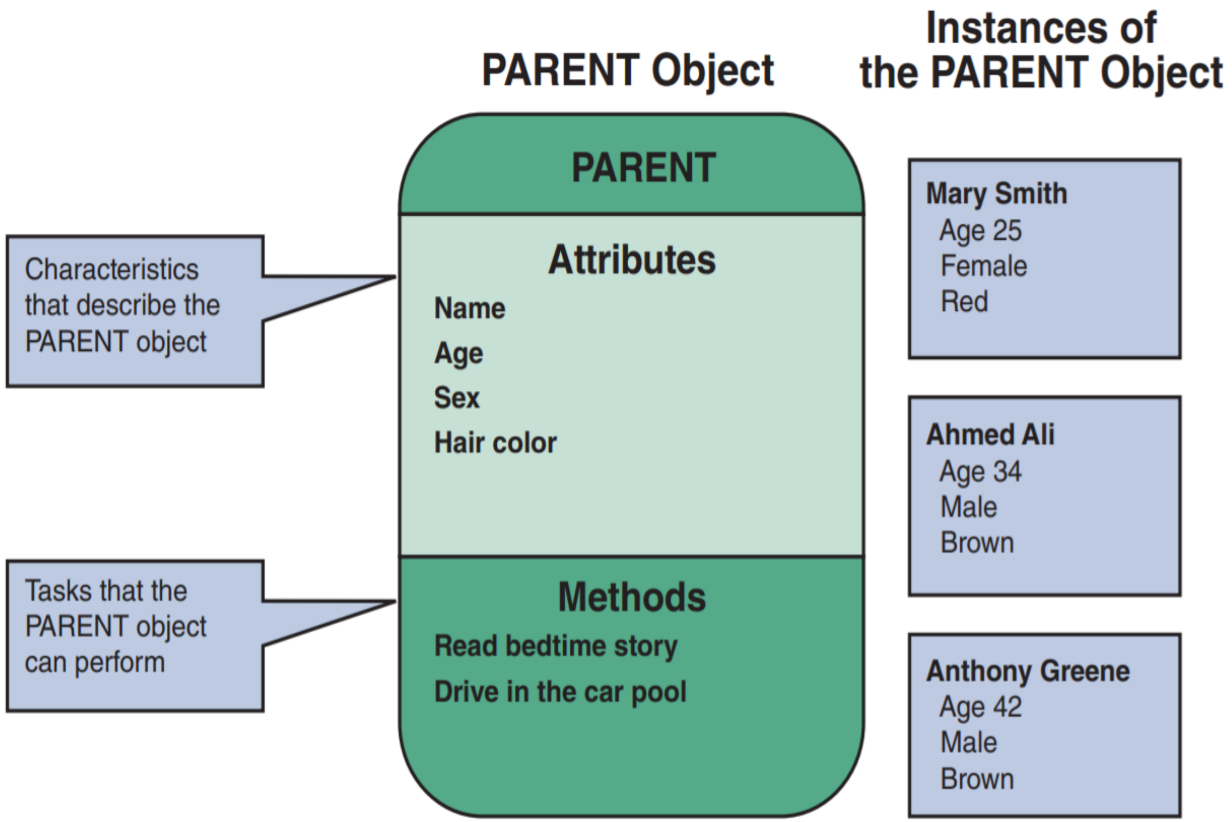
OBJECTS

- Consider a simplistic example of how the UML might describe a family with parents and children.
- UML represents an object as a rectangle with the object name at the top, followed by the object's attributes and methods.
- PARENT object with certain attributes such as name, age, sex, and hair color.
- If there are two parents, then there are two instances of the PARENT object.
- The PARENT object can perform methods, such as reading a bedtime story, driving the carpool van.



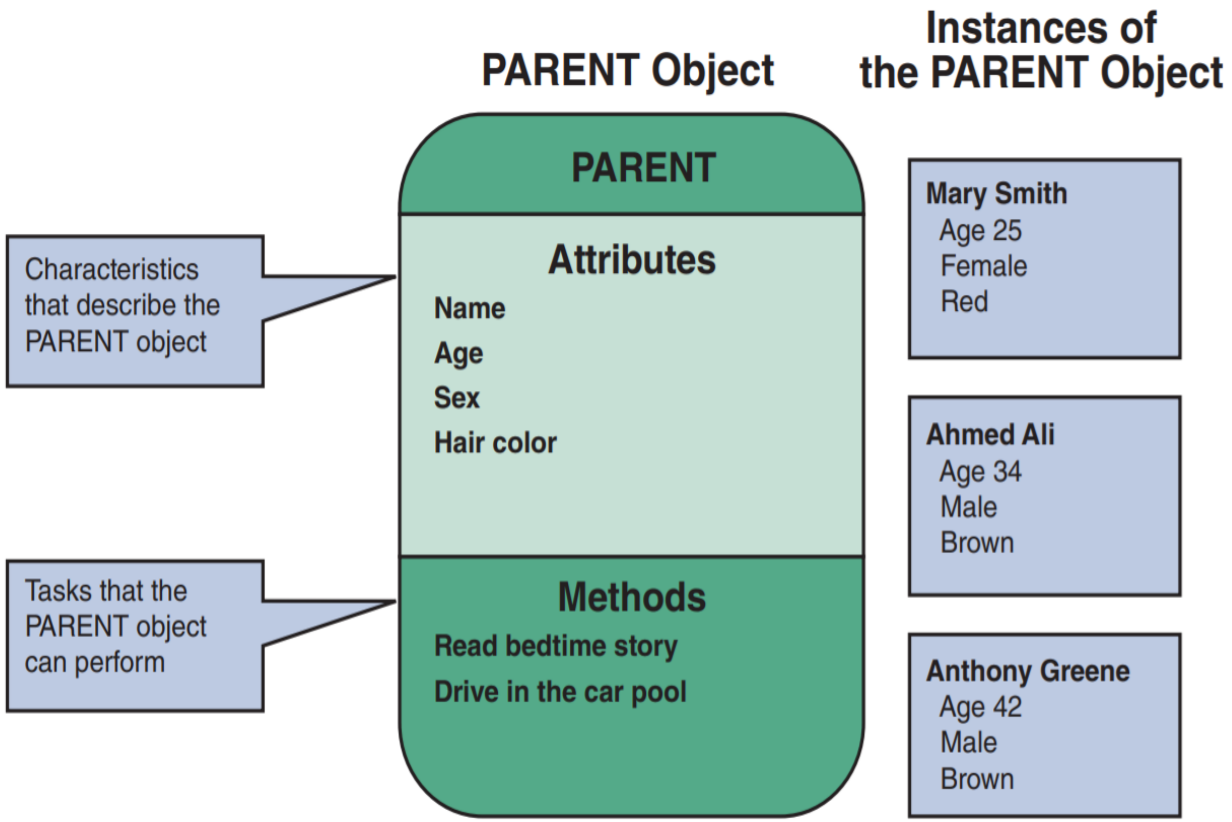
ATTRIBUTES

- If objects are similar to nouns, attributes are similar to adjectives that describe the characteristics of an object.
- How many attributes are needed?
- The answer depends on the business requirements of the information system and its users.



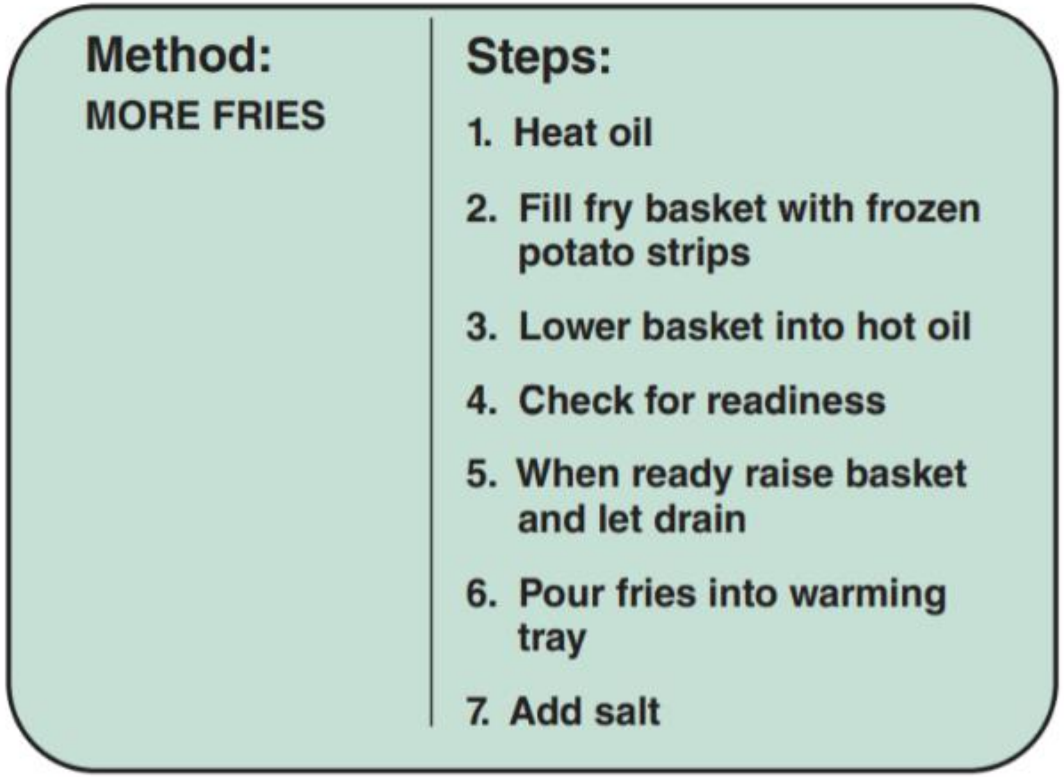
ATTRIBUTES

- Systems analysts define an object's attributes during the **systems design process**.
- In an O-O system, objects can inherit, or acquire, certain attributes from other objects.
- Objects can have a specific attribute called a **state**.
- The state of an object is an adjective that describes the object's current status.



METHODS

- A method defines specific **tasks** that an object can perform.
- Just as objects are similar to nouns and attributes are similar to adjectives, methods resemble verbs that describe what and how an object does something.



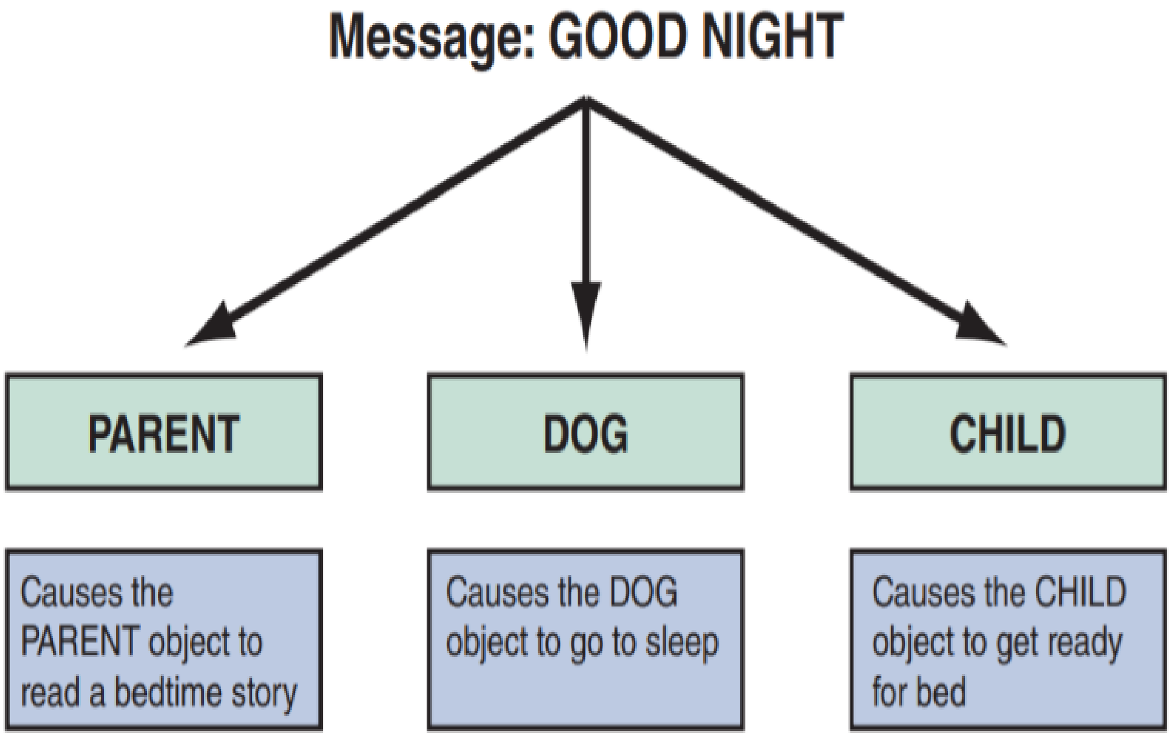
METHODS

- Consider a server who prepares fries in a fast-food restaurant.
- A systems analyst might describe the operation as a method called MORE FRIES.

Method:	Steps:
MORE FRIES	1. Heat oil 2. Fill fry basket with frozen potato strips 3. Lower basket into hot oil 4. Check for readiness 5. When ready raise basket and let drain 6. Pour fries into warming tray 7. Add salt

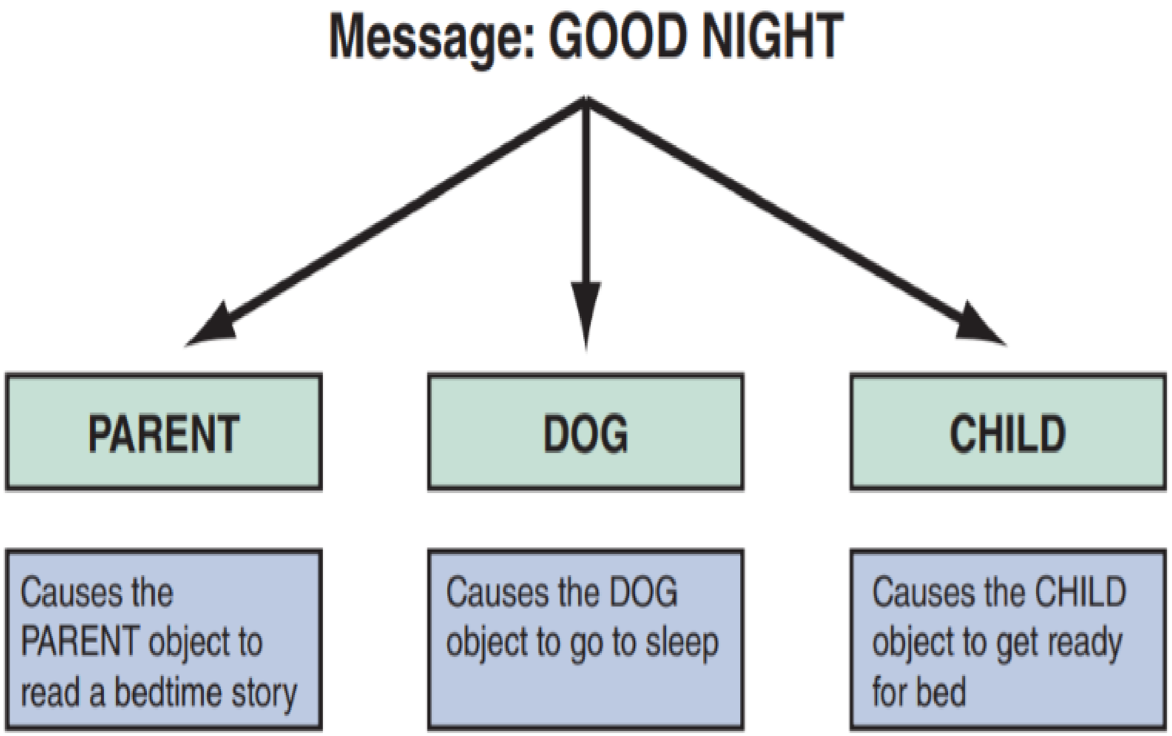
MESSAGES

- A message is a **command** that tells an object to perform a certain method.
- For example, the message PICK UP TOYS directs the CHILD class to perform all the necessary steps to pick up the toys.
- The CHILD class understands the message and executes the method.



Polymorphism

- The same message to two different objects can produce different results.
- The concept that a message gives different meanings to different objects is called **polymorphism**.
- For example, the message GOOD NIGHT signals the PARENT object to read a bedtime story, but the same message to the CHILD object signals it to get ready for bed.
- If the family also had a DOG object



Black Box

- An object can be viewed as a **black box**, because a message to the object triggers changes within the object without specifying how the changes must be carried out.
- A gas pump is an example of a black box. When the economy grade is selected at a pump, it is not necessary to think about how the pump determines the correct price and selects the right fuel, as long as it does so properly.

Encapsulation

- The black box concept is an example of **encapsulation**, which means that all data and methods are **self-contained.**

- A black box does not want or need outside interference.
- By limiting access to internal processes, an object prevents its internal code from being altered by another object or process.

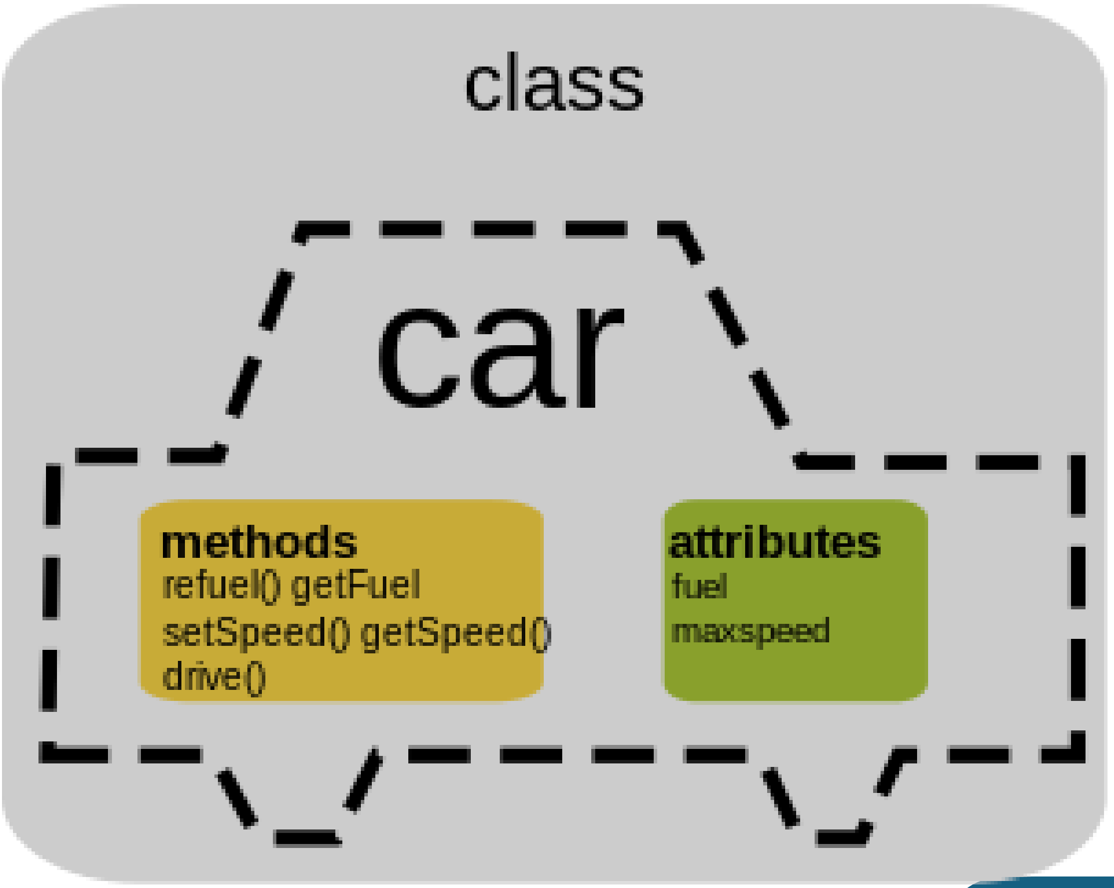
Encapsulation

- Encapsulation allows objects to be used as modular components anywhere in the system, because objects send and receive messages but do not alter the internal methods of other objects.

- The major advantages of O-O designs are that systems analysts **can save time and avoid errors** by using modular objects, and programmers can translate the designs into code, working with **reusable program modules** that have been tested and verified.

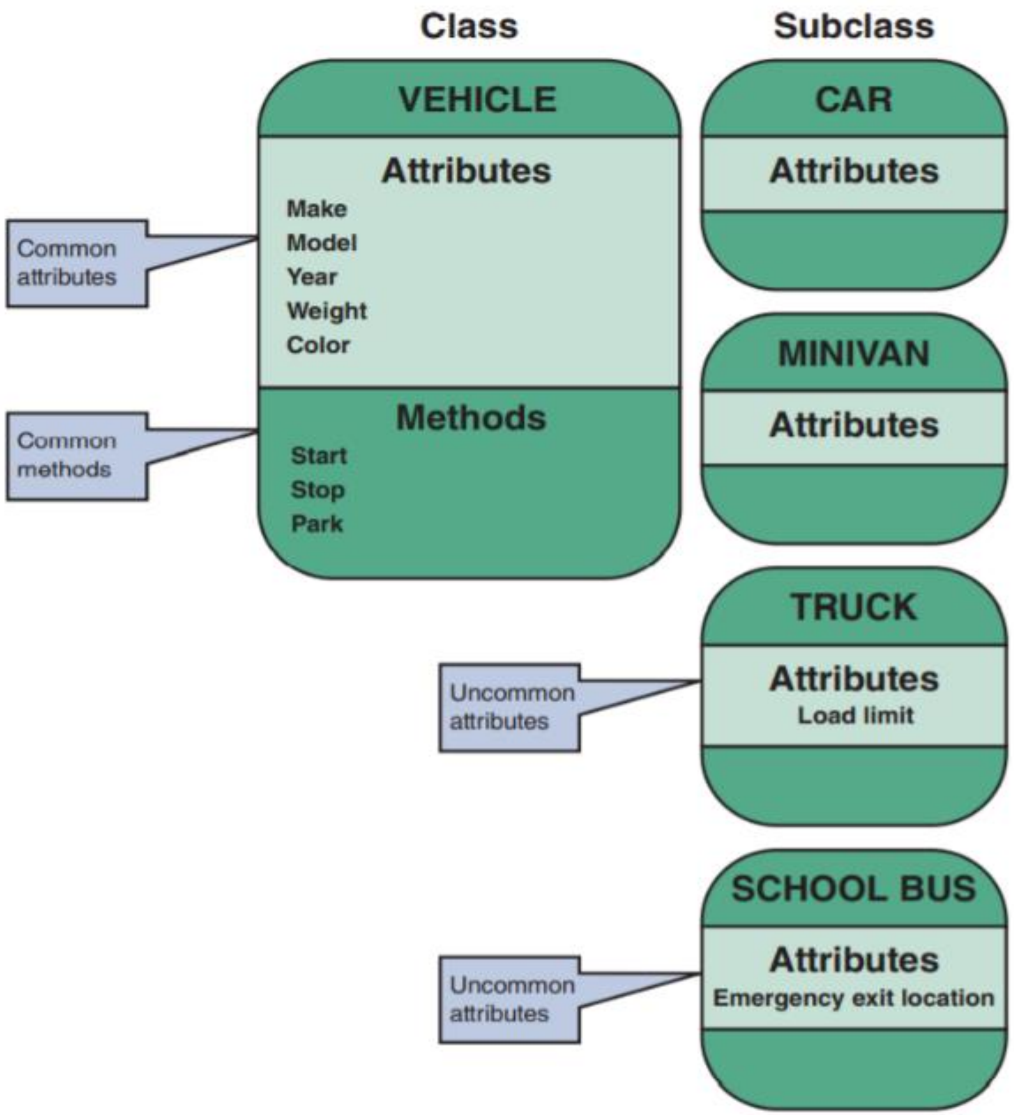
CLASSES

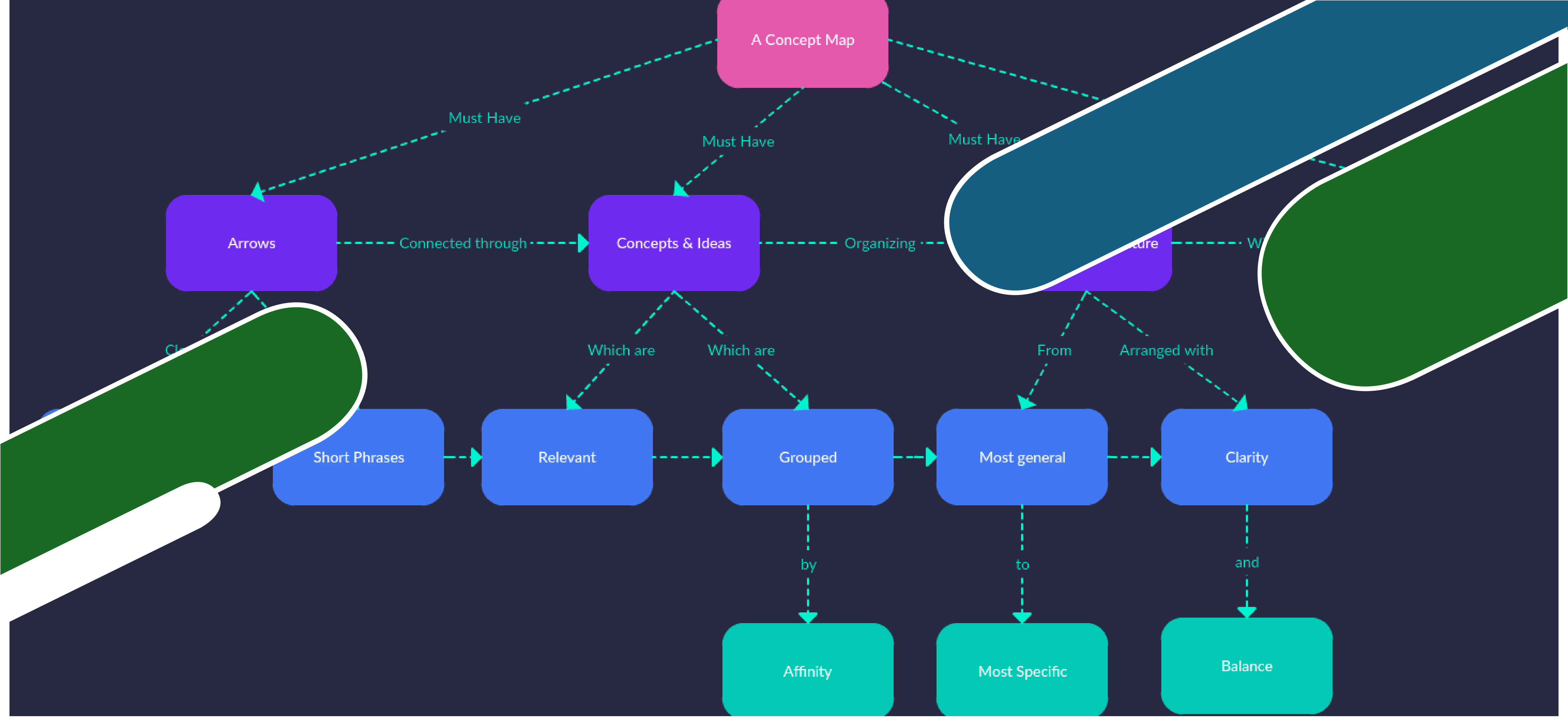
- An object belongs to a group or category called a **class**.
- All objects within a class share common attributes and methods, so a class is like a **blueprint**, or **template** for all the objects within the class.



CLASSES

- A class can belong to a more general category called a **superclass**.
- For example, a NOVEL class belongs to a superclass called BOOK, because all novels are books in this example. The NOVEL class can have subclasses called HARDCOVER, PAPERBACK, and DIGITAL.

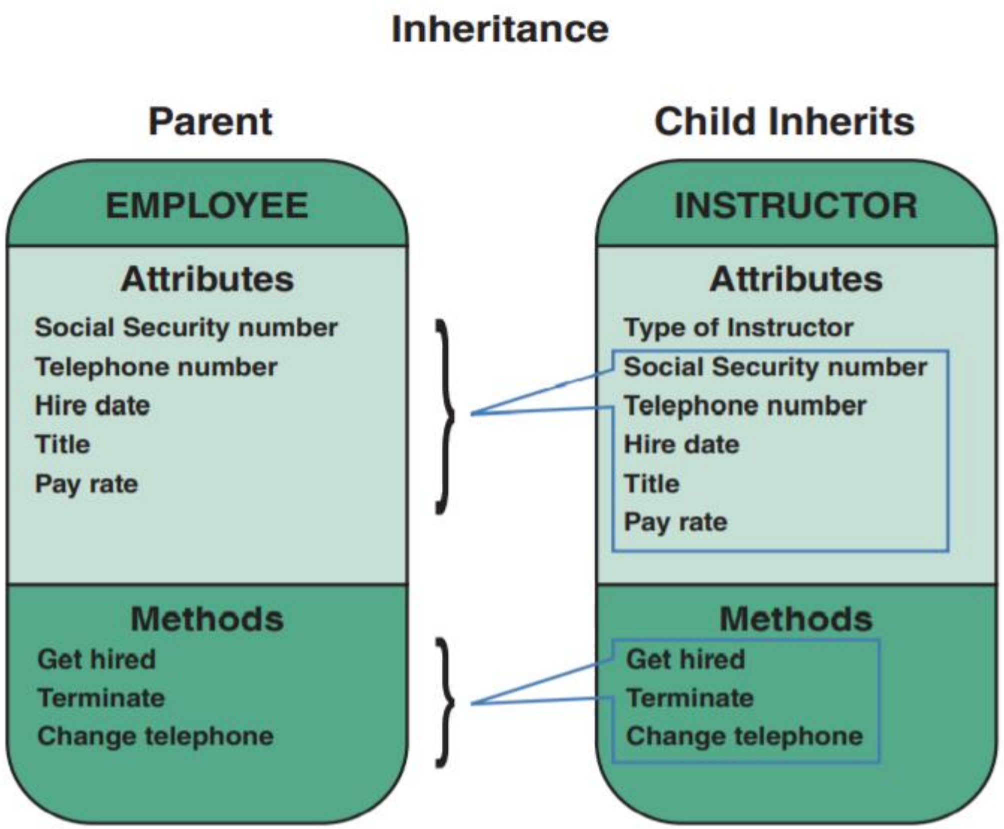




Relationship among objects and classes

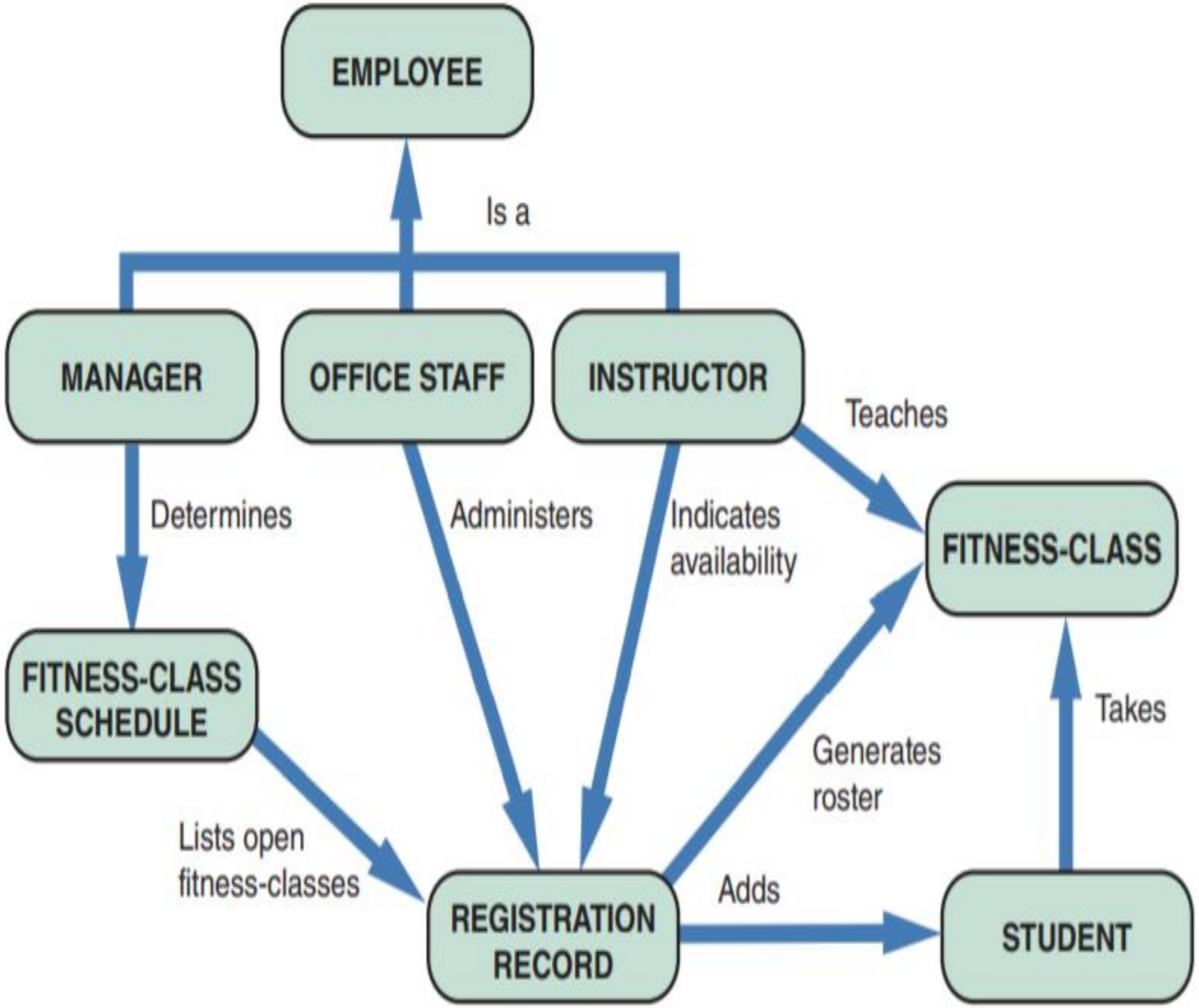
Relationships

- Relationships enable objects to **communicate** and **interact** as they perform business functions and transactions required by the system.
- Relationships describe what objects need to know about each other, how objects respond to changes in other objects, and the effects of membership in classes, superclasses, and subclasses.



OBJECT RELATIONSHIP DIAGRAM

- An object relationship diagram can be prepared to provide an overview of the system.
- That model is used as a guide to continue to develop additional diagrams and documentation.



THANK YOU!